

Executive Summary: Filipino Complex Emotions & Affective Analysis Through¹ LLMs

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Introduction & Motivation

The study addresses the challenges of emotion classification and affective analysis in Filipino, an under-resourced language in natural language processing (NLP). While English emotion classifiers using large language models (LLMs) have shown promising results, their performance in Filipino remains underexplored due to the language's unique morphological and syntactic structures. This research builds on prior works, including Tumaliuan et al. (2024), Montalan et al. (2025), and Pippin et al. (2015), to evaluate the effectiveness of LLM-based English emotion classifiers in Filipino and develop a corpus of Filipino statements for further affective analysis.

Objectives

1. Build a corpus of Filipino statements reflecting complex emotions to serve as a baseline for future NLP research.
2. Train and evaluate LLMs to classify Filipino statements into seven affective states: sadness, joy, love, anger, fear, anxiety, and stress.

Methodology

Data Preparation

The study consolidated an English emotion dataset from Pandey (2021) and Balatska (2024), comprising 59,210 phrases categorized into seven emotions. The dataset was translated into Filipino using machine translation models, including Gemma-SEA-LION-v3-9B-IT. Translations were evaluated using MetricX-24 (Montalan et al., 2025) and Metric-24-hybrid-xl-v2p6 (Google), with a threshold of 87.24 for quality assurance. Human evaluation showed a 94.4% success rate for translation accuracy.

Data Augmentation

To address the imbalance in the dataset, dominant-class (i.e. sadness) downsampling and data augmentation techniques were applied, including (as recommended by Pippin et al, 2015):

- Taglish (mix of Tagalog and English)
- Jejemon (text with unconventional spelling)
- Slang
- Vowel stripping
- Erroneous spelling
- Paraphrasing

The final augmented dataset consisted of 32,162 Filipino phrases, enriched with Filipino-specific linguistic nuances.

¹This executive summary was drafted with the aid of Adobe Acrobat AI Assistant.

Model Training

Four pre-trained LLMs were tested for emotion classification:

1. **jcblaise/roberta-tagalog-base** (Cruz & Cheng, 2021)
2. **jcblaise/roberta-tagalog-large** (Cruz & Cheng, 2021)
3. **xlm-roberta-base** (Conneau et al., 2019)
4. **bert-base-multilingual-cased** (Devlin et al., 2018)

Models were trained using both fully frozen and partially frozen tail configurations. The best-performing model was **jcblaise/roberta-tagalog-large** with partially frozen tails, achieving an accuracy of 75.59%, precision of 74.12%, recall of 74.06%, and F1-score of 74.06%. These results are better than Pippin et. al (2015).

Results & Analysis

Corpus Generation

The original dataset was imbalanced, with sadness being the dominant emotion (53%). After downsampling and augmentation, the dataset achieved a more balanced distribution across the seven emotions. The augmented dataset included 32,162 Filipino phrases, incorporating diverse linguistic styles such as Taglish, jejemon, slang, and vowel-stripping.

Model Performance

The study demonstrated that LLMs could effectively classify Filipino text into complex emotions. The Roberta-Tagalog-Large model outperformed other models, showcasing its suitability for Filipino emotion classification tasks.

Key Findings

- The study successfully created a Filipino emotion dataset with 32,162 phrases, enriched with linguistic nuances unique to Filipino.
- Machine translation achieved a 94.4% human-evaluation success rate in translating English emotion phrases to Filipino.
- The Roberta-Tagalog-Large model achieved the highest performance in classifying Filipino emotions, with an accuracy of 75.59%.

Challenges

1. Limited manual evaluation of generated texts due to time constraints.
2. Character limits in text translation led to truncation of some Filipino texts.
3. GPU availability restricted the scope of model training.
4. Insufficient data for undersampled emotion categories (e.g. love).

Conclusion

This study provides a foundational framework for Filipino emotion classification using LLMs. The generated Filipino emotion dataset and the insights from model evaluation can serve as valuable resources for future research in affective analysis and NLP for under-resourced languages. The findings build on prior works, including Tumaliuan et al. (2024), Montalan et al. (2025), and Pippin et al. (2015), and highlight the potential of LLMs in addressing linguistic challenges in Filipino NLP.